Om Amit Gandhi

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Objective

Seeking distributed systems / blockchain infrastructure internships focused on scalable protocol design and performance. Interested in building simulators and experiments to evaluate throughput, latency, propagation, and protocol tradeoffs.

Research Interests

Blockchain sharding; scalable consensus; network-aware execution and propagation. Performance modeling, measurement, and reproducible experimental workflows at scale.

Education

2025–Present	**Ph.D. Student in Computer Science**, Illinois Institute of Technology	(GPA: 3.93/4.0)
2024–2025	**Masters in Computer Science**, Illinois Institute of Technology	(GPA: 3.9/4.0)
2019–2023	**B.E. in Computer Engineering**, Gujarat Technological University	(GPA: 3.6/4.0)

Experience

Research Assistant | Illinois Institute of Technology May 2025 – Present

- Conduct research in blockchain sharding and distributed systems, focusing on scalability and performance tradeoffs.
- Develop simulation and experimental workflows to evaluate throughput, latency, and protocol behavior.
- Summarize findings through plots/tables and iterative reporting to support research progress and publications.

Teaching Assistant | Illinois Institute of Technology Aug 2025 – Present

- Support students through office hours, troubleshooting, and structured feedback on assignments and core course concepts.
- Courses assisted: CS458 (Aug 2025 – May 2026); CS350, CS528 (Aug 2026 – Present).

Junior Web Developer | Minimal Dot Feb 2023 – Jul 2023

- Built and shipped client-facing UI features and bug fixes using Angular.

Web Development Intern | TatvaSoft Jun 2022 – Jul 2022

- Contributed to a team e-commerce web app; implemented UI flows using TypeScript (with exposure to PostgreSQL/.NET).

Research Projects

Blockchain Sharding Simulator (Discrete-Event Simulation) May 2025 – Present

- Built a simulator to compare sharded vs. non-sharded execution under varied network/workload settings.
- Modeled nodes/miners and propagation (latency/bandwidth); measured throughput and confirmation latency.
- Ran parameter sweeps and summarized results with plots/tables for research iteration.

Publications

- Om Amit Gandhi, Ioan Raicu. **The Cost of Coordination: A Discrete-Event Simulation Study of Throughput Saturation in Sharded Blockchains**, under review at IEEE MASCOTS 2026.
- Om Amit Gandhi, Ioan Raicu. **Simulation-Based Performance Evaluation of Sharded Blockchain Architectures**. arXiv:2606.14394.
- Samuel T. Fatunmbi, Om Amit Gandhi, Luke Logan. **VaultxGPU: GPU-Accelerated Blockchain Consensus**. arXiv:2606.14007.

Presentations

- Real System Benchmark: Simulation Validation — DataSys Lab, Illinois Institute of Technology May 2026
- Simulation-Based Performance Evaluation of Sharded Blockchain Architectures — Northwestern University May 2026
- Shards on a Shoestring: Putting Blockchain Sharding Simulations to the Test — Northwestern University May 2026

Skills

Languages	C, C++, Python, Scala
Frameworks/Tools	React, Angular, Flask, Node.js, Flutter, SimPy, OpenMP, Docker
Systems	Linux, Git, VS Code, Vim, distributed systems, blockchain sharding
Writing	LaTeX, Overleaf